ETHERNET STANDARD AND SPECIFICATION

Course: Diploma in Cyber Security

Topic: Ethernet Standard and Specification

Name: HET ACHARYA

Introduction

Ethernet standards describe how wired network devices communicate. They also give information about speed, cable type and the connection. In this assignment, I have covered T568A and T568B wiring, common Ethernet standards and direct laptop connections.

Task 1: RJ-45 Pin Layout and T568A / T568B

An RJ-45 Ethernet connector has 8 pins. T568A and T568B use the same eight wires but arrange the pairs in a different order.

Pin	T568A	T568B
1	White-Green	White-Orange
2	Green	Orange
3	White-Orange	White-Green
4	Blue	Blue
5	White-Blue	White-Blue
6	Orange	Green
7	White-Brown	White-Brown
8	Brown	Brown

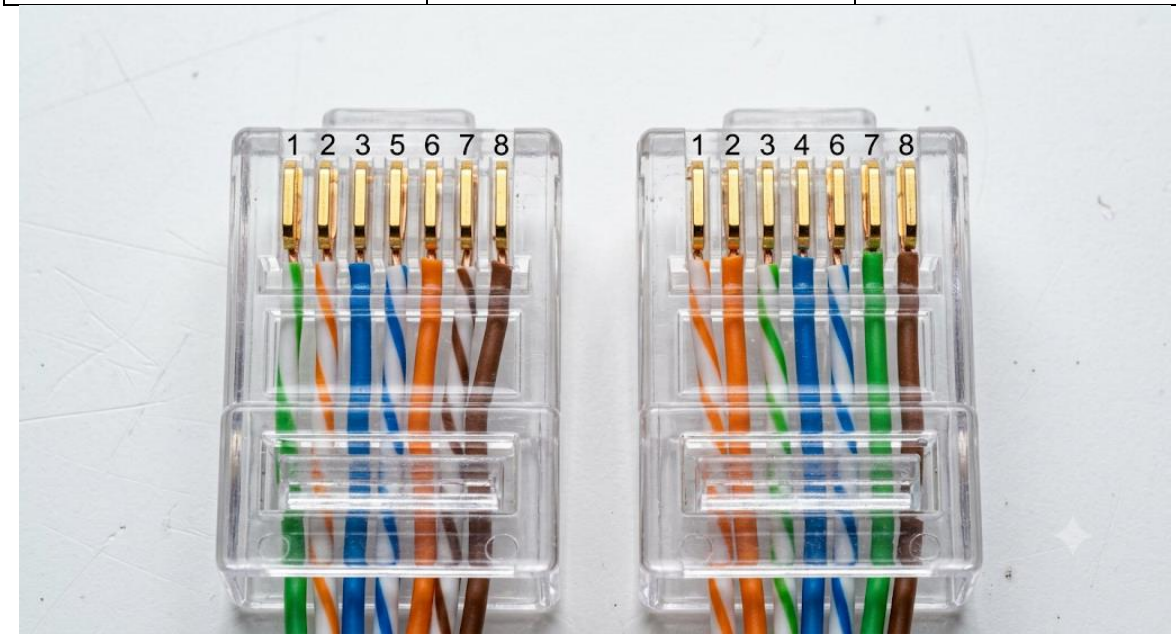


Figure 1: RJ-45 Pin Layout with T568A and T568B

Task 2: Ethernet Standards Comparison

Standard	Speed	Cable Type	Maximum Length	Connector
10BASE-T	10 Mbps	Twisted-pair Ethernet cable	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better twisted pair	100 m	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e or better twisted pair	100 m	RJ-45

Task 3: Connecting Two Laptops Directly

For directly connecting two laptops with Ethernet, a crossover cable can be used when the network ports do not automatically correct the transmit and receive pair arrangement. The crossover cable swaps the relevant transmit and receive pairs so similar devices can communicate directly. Many modern Ethernet ports support Auto-MDI/MDI-X, so a normal straight-through cable may also work.

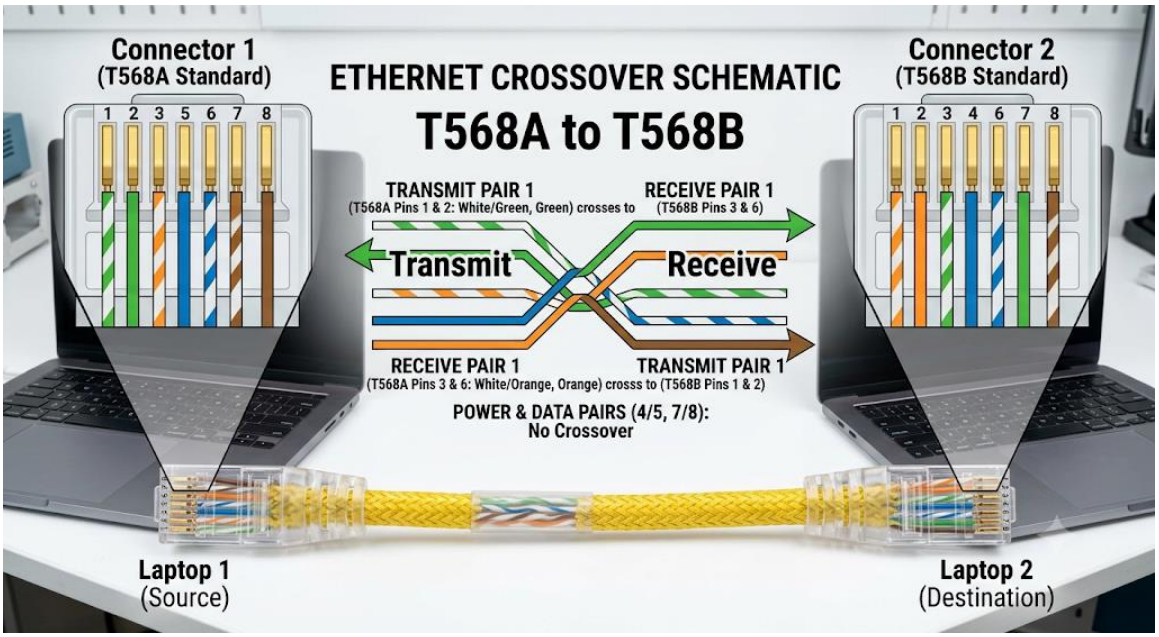


FIGURE 2: Draw Laptop 1 -> Ethernet Crossover Cable -> Laptop 2, and show the crossed transmit/receive pairs.

Task 4: Devices Using Ethernet Connections

Device	Likely Ethernet Standard	Use
College desktop PC	1000BASE-T	Connects to the college LAN for internet and internal services.
Wi-Fi Router / Network Switch	1000BASE-T on suitable LAN ports	Connects wired devices and transfers network data.
Smart TV / Gaming Console	1000BASE-T or 100BASE-TX depending on	Used for streaming, updates and

	model	**online gaming.**

Conclusion

In this assignment, I learned about RJ-45 pins, T568A and T568B wiring and common Ethernet standards. I also understood why cable type and wiring matter when connecting devices.

Quick Revision

- RJ-45 Ethernet connector has 8 pins.
- T568A and T568B are two wiring standards.
- 10BASE-T = 10 Mbps.----
- 100BASE-TX = 100 Mbps.
- 1000BASE-T = 1 Gbps.
- Crossover cable can be used for direct connection of similar devices when Auto-MDI/MDI-X is not available.